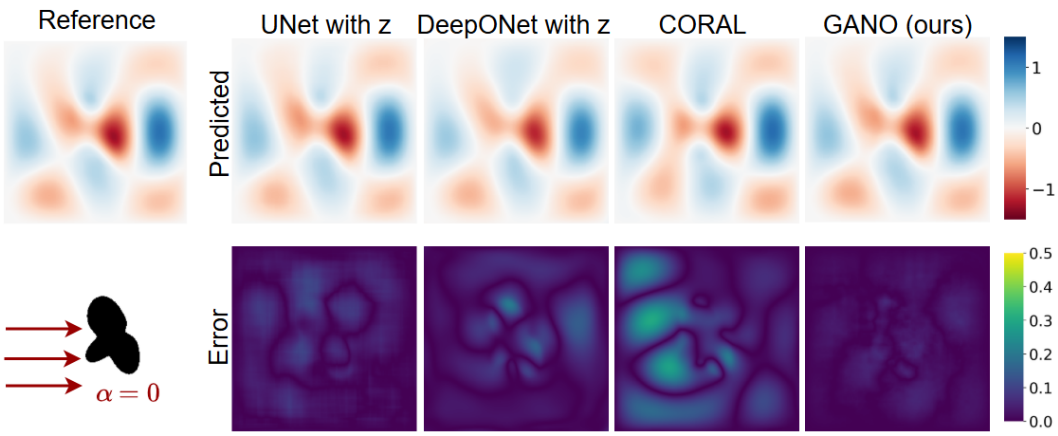
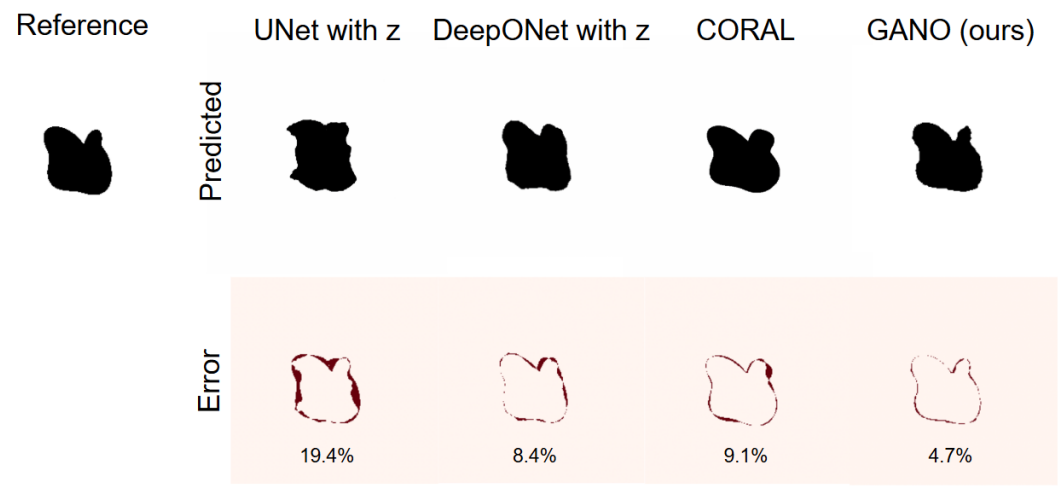
Table 1: **Projection success rates under different absolute SDF thresholds.** The statistics are computed on 100 vehicles, each with 10^5 query points, resulting in 10^7 points in total. A point is considered successfully projected if its absolute SDF value is below the threshold.

Threshold	Before projection	After projection
10^{-6}	0.0024%	98.6350%
10^{-5}	0.0231%	99.7322%
10^{-4}	0.2281%	99.9739%
10^{-3}	2.2684%	99.9998%



(a)



(b)

Figure 1: **2D Helmholtz tasks: comparison of geometry-injected UNet and DeepONet, CORAL, and GANO.** (a) Forward task. In CORAL, Fourier shape parameters are first used to generate the obstacle boundary, and the region between this boundary and a fixed outer boundary is then interpolated into a 2-channel coordinate field as the geometric input encoded by CORAL. When predicting the latent physical field, the geometric latent and the incident angle are concatenated as the input to the MLP. Other settings are consistent with its open-source code. (b) Inversion task.

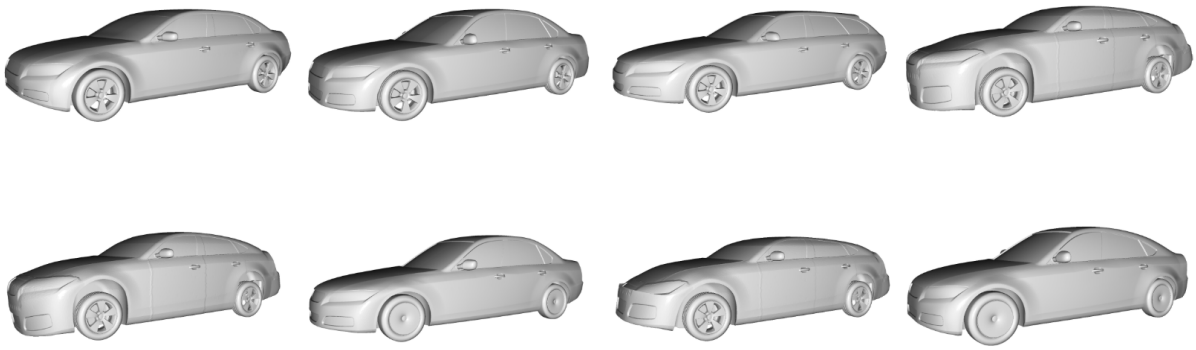


Figure 2: **Different categories of vehicle types in DrivAerNet++.**

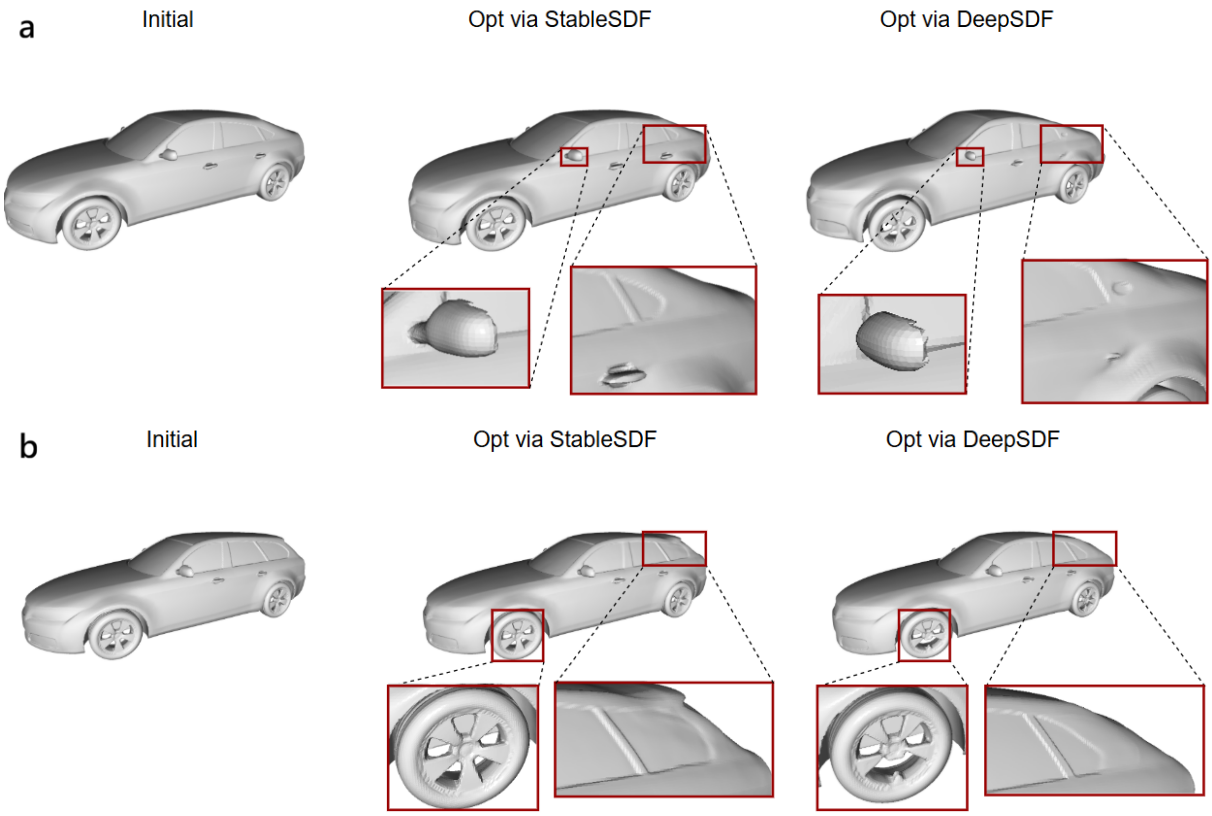


Figure 3: **Comparison of the results after optimizing the same Fastback and Estateback car for 20 steps using StableSDF and DeepSDF under the same settings.** (a) After optimization with DeepSDF, the deformation of the car details is relatively severe: the side mirrors detach from the car body, and the rear door handle as well as the rear window also undergo severe deformation. (b) After optimization with DeepSDF, the wheels show obvious damage, and the rear back of the car deviates severely from the original design, changing from an Estateback into a Fastback.

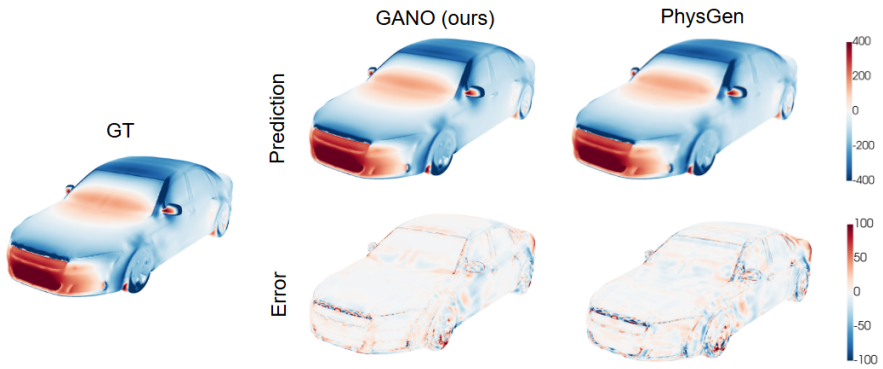


Figure 4: **Comparison of the results of GANO and PhysGen in predicting the pressure field.**

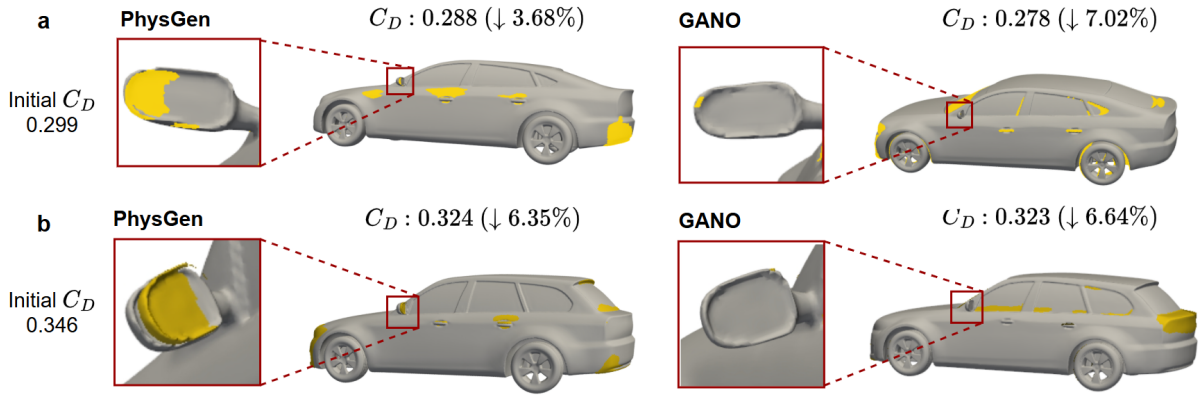


Figure 5: **Comparison of GANO and PhysGen on vehicle optimization.** Gray denotes the input vehicle, and yellow highlights regions with large geometric changes after optimization. In (a), GANO achieves a better result than PhysGen; in (b), the two are comparable. However, PhysGen often reduces drag by shrinking or sweeping back the side mirrors, while GANO largely preserves them due to null-space projection while still achieving meaningful drag reduction. The enlarged views highlight this difference.

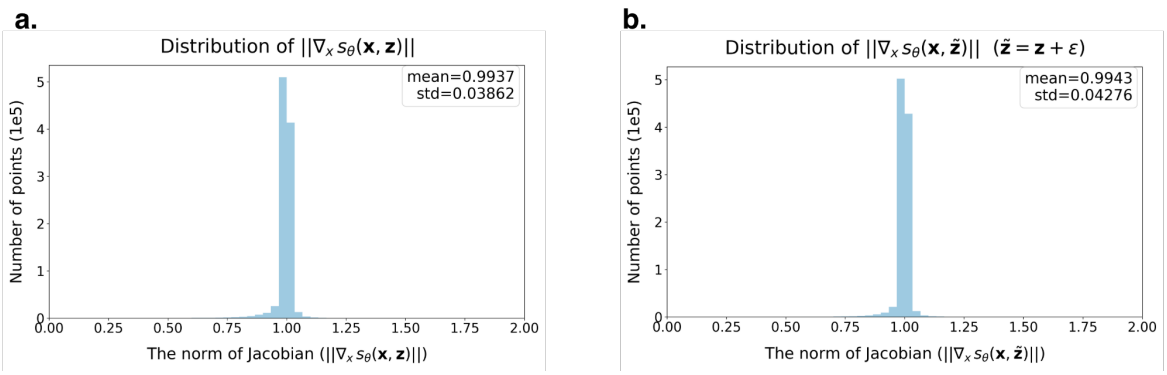


Figure 6: **Distribution of the Jacobian norm of the StableSDF output with respect to x:** (a) before random perturbation and (b) after random perturbation. The statistics are computed from 100 randomly sampled test vehicles, with 10,000 points sampled from each vehicle.

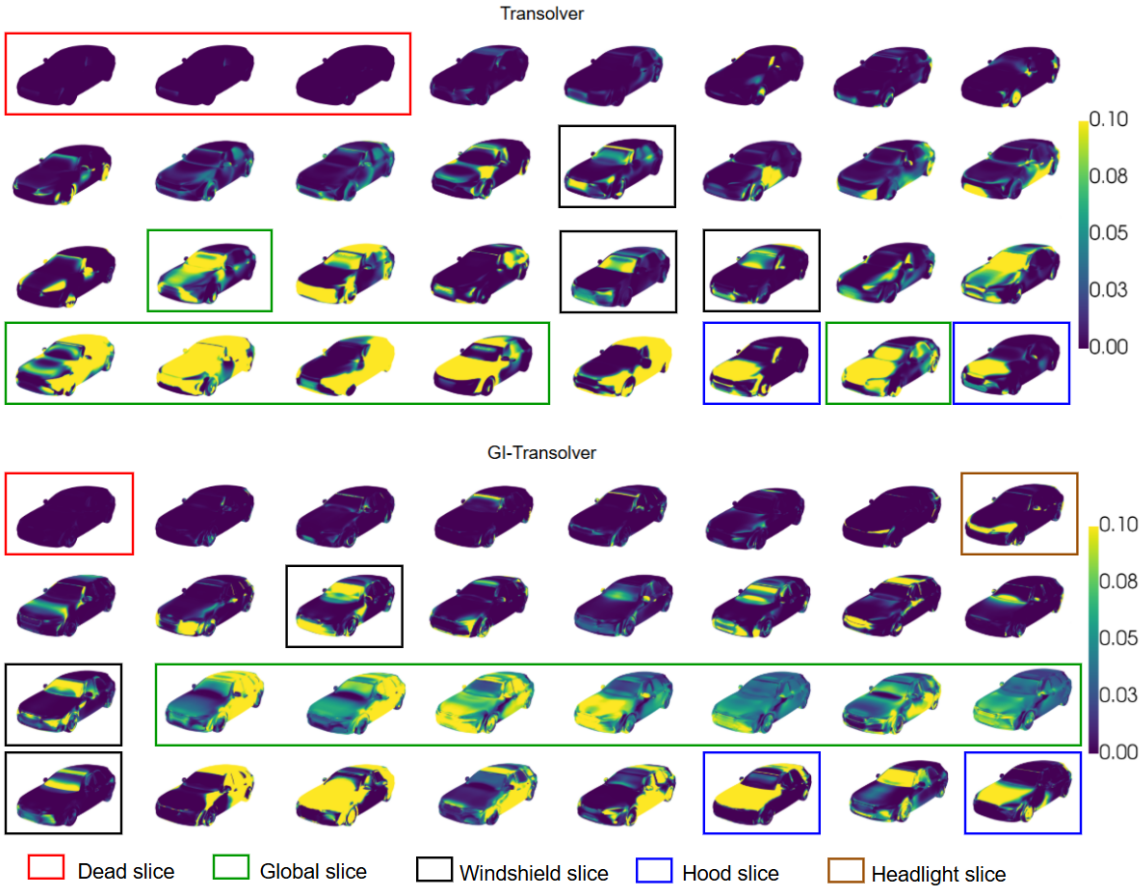


Figure 7: **Comparison of slices of GI-Transolver and Transolver.** 1) Red indicates dead slices. A dead slice refers to a slice with very low activation values and no clear semantic meaning. GI-Transolver has fewer dead slices than Transolver. 2) Green indicates global slices. A global slice has relatively high activation values over the whole object. Compared with localized activations, such globally high activations can be interpreted as corresponding to a global physical feature. Due to our colorbar setting, the visual differences within the range from 0.03 to 0.08 are not very pronounced, so the slice may appear blurry for GI-Transolver. 3) Black indicates the windshield slice. Both models capture this region. 4) Blue indicates the hood slice. The hood region learned by our model is more complete than that learned by Transolver. 5) Brown indicates the headlight slice. In contrast to ours, Transolver shows a less localized pattern on this region.